# Lecture 11: Demand estimation

ECON 7510

Cornell University

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Slides draw upon Berry and Haile (2021) and upon lecture materials from Nikhil Agarwal (MIT).

# Introduction to Parts 3-4 of the course

In this part of the course, we'll cover two topics fundamental to structural empirical work:

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- What can we identify from data?
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Key principles to keep in mind:

- Demand estimation / DDC estimation are not *ends* in and of themselves. They are *means* of answering economic questions.
- In answering the modeling/methodology questions above, ask yourself: What are the salient features of my setting? What do I want to do with my demand/DDC estimates?

# Demand

- As we've seen, the shape of a demand curve determines answer to questions about, e.g., price discrimination, merger analysis, etc.
- In our theoretical work so far, we've taken as given the demand curve $D(p)$.
- In empirical applications, we need to *estimate* it from data.

# Data

**What kind of data might we have?**

1. **Consumer-level** data
   - Individuals' purchases: The goods they buy and the prices they pay. Maybe as a panel.
   - Often some consumer characteristics.
2. **Market-level** (aggregate) data
   - For each market (e.g., year or region $\times$ month), observe prices and total quantities sold.
3. Market-level data + some limited consumer-level demographic data. For example,
   - Second-choice survey data.
   - Demographic info (e.g., the average income of a BMW purchaser is $225,000).

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# Outline

1. Challenges in demand estimation
2. Introduction to demand models
3. Overview of estimation
4. Identification
5. Instruments
6. Incorporating the supply side
7. Implementing the estimator
8. Applications
   - 8.1 Demand for cars (BLP)
   - 8.2 “Micro BLP”; welfare effects of introducing the minivan (Petrin)
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2. Demand for one good depends on every other good's price/characteristics/**structural errors**.

   → Demand is not regression!

$$Q_j = D_j(X, P, U)$$

where $j \in \{1, \ldots, J\}$ indexes products, $X = (X_1, \ldots, X_J)$ are product characteristics, $P = (P_1, \ldots, P_J)$ are product prices, and $U = (U_1, \ldots, U_J)$ are demand shocks.

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**Approach**: To tackle the second challenge, we draw on literature on simultaneous equations ("inversion"). Then, to tackle the first, we can use more familiar IV tools.

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→ Experimental variation in prices is insufficient to identify demand.

# Why won't a simple solution work?

## *Fixed effects*

Suppose we introduce fixed effects to control for the demand shocks.

- Product $\times$ market ($j \times t$)-level fixed effects.

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  **Problem**: If assumption is correct, model should fit data perfectly. If fit is not perfect, then there is still a $jt$ error floating around....

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2. Model utility, not demand.
   - Easier to understand what economic restrictions we're imposing.
   - A surprising benefit: Modeling utility may actually be more practical; can model a complicated demand system with relatively few parameters.
     - » Moreover, if we estimate demand rather than utility, we may be limited in counterfactuals we can evaluate.

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    - **The canonical RUM**: A specific parameterization. (McFadden 1981, Berry 1994, BLP 1995, and others).
    - **Vertical models**—products only vertically differentiated; consumers only differentiated in their taste for quality.

# An illustrative example

*A model of vertical differentiation (Bresnehan 1987)*

- Vertical model:

$$u_{ij} = \overbrace{(x_j\beta + \xi_j)}^{\delta_j}\nu_i - p_j$$

ordered so that $\delta_0 < \delta_1 < \ldots < \delta_J$. Here, $\delta_j$ is measure of quality and $\nu_i$ is how much consumer $i$ cares about quality (relative to price).

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  - Choose $j$ if $u_{ij} > u_{ij-1}$ and $u_{ij} > u_{ij+1}$.
  - If $F_\nu$ is the CDF of $\nu$ (assumed, for the moment), then

$$s_j = F_\nu\left(\frac{p_{j+1} - p_j}{\delta_{j+1} - \delta_j}\right) - F_\nu\left(\frac{p_j - p_{j-1}}{\delta_j - \delta_{j-1}}\right)$$

- Solve ("invert") the system to get $\delta = S^{-1}(\hat{s}; x, p, \theta)$, where $\hat{s}$ are observed market shares
  - Based on assuming many consumers: $\frac{N}{J} \to \infty$, so that

$$\hat{s} - s \to O_p(1/\sqrt{N})$$

allowing us to ignore sampling noise in aggregate market shares

# Random utility models

- Discrete choice models for consumers $i$ with unit demand for products $j \in \{0, \ldots, J\}$
- Indirect utility for products

$$U_{ij} = U(x_j, p_j, \xi_j, \varepsilon_i)$$

where

| | |
|---|---|
| $\varepsilon_i$ | : unobserved consumer characteristics (taste shifters) |
| $x_j, \xi_j$ | : observed and unobserved product characteristics |
| $p_j$ | : price |

- ✓ Consumers pick

$$\arg \max_{j \in \{0,\ldots,J\}} U_{ij}$$

  - Typically estimated with parametric assumptions
  - Typically call $j = 0$ the outside option and normalize $U_{i0}$. (More on this in a second.)

# Random utility models: Choice probabilities

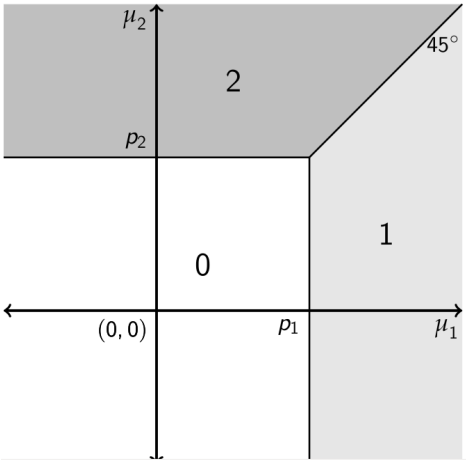
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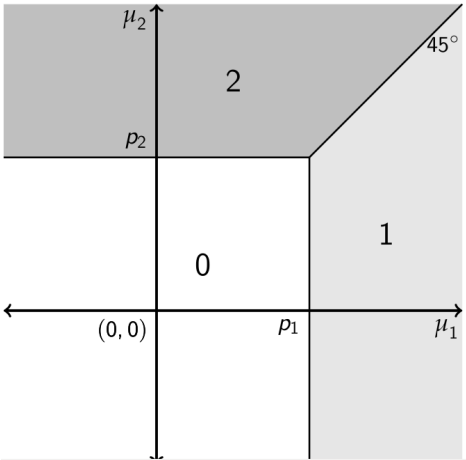



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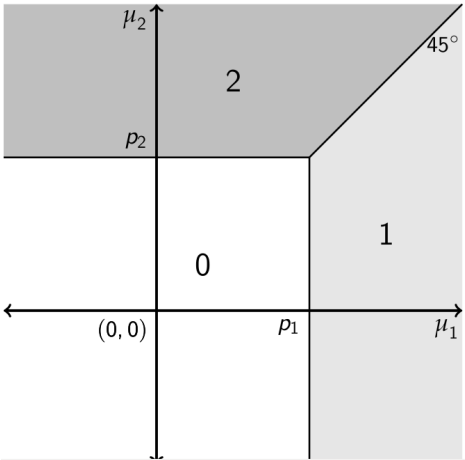



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- **Question**: In this figure, how do we think about the effects of increasing $p_1$?
- **Question**: Looking at this figure, is it possible that there is substitution between 0 and 1 and between 1 and 2, but no substitution between 0 and 2?

# The canonical model

A parametric specification:

$$u_{ijt} = \begin{cases} x_{jt}\beta_{it} - \alpha_{it}p_{jt} + \xi_{jt} + \varepsilon_{ijt} & \text{if} \quad j > 0 \\ \varepsilon_{i0t} & \text{if} \quad j = 0 \end{cases}$$

where

$$\begin{aligned} \varepsilon_{ijt} &: \text{unobserved consumer characteristics (taste shifters)} \\ x_{jt} \in \mathbb{R}^K, \xi_{jt} \in \mathbb{R} &: \text{observed and unobserved product characteristics} \\ p_{jt} &: \text{price} \end{aligned}$$

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- If $\beta_{it} = \beta$ and $\alpha_{it} = \alpha$ for all $i$, this is *multinomial logit model*.
- Otherwise, it's a kind of mixture model: *mixed logit*, a.k.a, *random coefficients logit*.

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*What does the T1EV assumption buy us?*

For a given $\beta_i, \alpha_i$,

$$s_{ijt} \equiv \Pr\left(a_{it} = j \mid x_t, p_t, \xi_t, \beta_{it}, \alpha_{it}\right) = \frac{\exp\left(x_{jt}\beta_{it} - \alpha_{it}p_{jt} + \xi_{jt}\right)}{1 + \sum_{j'} \exp\left(x_{j't}\beta_{it} - \alpha_{it}p_{j't} + \xi_{j't}\right)}$$

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$$\begin{aligned} s_{jt} &\equiv \int \Pr\left(a_{it} = j \mid x_t, p_t, \xi_t, \beta_{it}, \alpha_{it}\right) dF(\alpha_{it}, \beta_{it}; t) \\ &= \int \frac{\exp\left(x_{jt}\beta_{it} - \alpha_{it}p_{jt} + \xi_{jt}\right)}{1 + \sum_{j'} \exp\left(x_{j't}\beta_{it} - \alpha_{it}p_{j't} + \xi_{j't}\right)} dF(\alpha_{it}, \beta_{it}; t) \end{aligned}$$

# The canonical model

*The distribution of random coefficients*

For each characteristic $k$, let

$$\beta_{it}^{(k)} = \beta_0^{(k)} + \beta_\nu^{(k)} \nu_{it}^{(k)} \underbrace{+ \sum_\ell \beta_d^{(\ell,k)} d_{i\ell t}}_{\text{if we have demographic data}}$$

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and, for example,

$$\log(\alpha_{it}) = \alpha_0 + \alpha_\nu \nu_{it}^{(0)} \underbrace{+ \alpha_y y_{it}}_{\text{if we have income/other } \alpha\text{-relevant data}}$$

Assume parametric distribution for $\nu_{it}$. A common choice is $\nu_{it} \sim N(0, \Sigma)$.

# Taking a step back

The canonical model features two kinds of randomness:

1. Random taste for products, $\epsilon_{ijt}$.
2. Random taste for product characteristics, $(\beta_{it}, \alpha_{it})$.

The second seems very complicated. What if we just keep the first (i.e., the logit model)?

# With the logit model, estimation is easy...

$$u_{ijt} = \begin{cases} \underbrace{x_{jt}\beta - \alpha p_{jt} + \xi_{jt}}_{\delta_{jt}} + \varepsilon_{ijt} & \text{if} \quad j > 0 \\ \varepsilon_{i0t} & \text{if} \quad j = 0 \end{cases}$$

Choice probabilities the same across all consumers, so market share is

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Then, how can we estimate $\beta$, $\alpha$? OLS (if no price endogeneity) or IV regression of $\delta_{jt}$ on $x_{jt}$, $p_{jt}$.

# … but logit imposes very strong *a priori* restrictions

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**Random coefficients help us avoid these issues!** (Nested logit—a special case of random coefficients logit—would be a smaller step away from logit.)

# So back to mixed logit. How do we estimate it?

Typical "market-level" data:

- Number of goods $J_t$,
- prices $p_t$ and observable characteristics $x_t$,
- market shares $\tilde{s}_{jt}$,
- distribution of characteristics $(d_{it}, y_{it})$ for each market $t$, and
- instruments $w_t$ (e.g., cost shifters).

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# Estimating mixed logit

For expositional simplicity, assume $\alpha_{it} = \alpha_0$; with random coefficient on price, approach would be very similar.

Write utility as

$$u_{ijt} = \delta_{jt} + \mu_{ijt} + \epsilon_{ijt}$$

where

$$\delta_{jt} = x_{jt}\beta_0 - \alpha_0 p_{jt} + \xi_{jt}$$
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Can then write market shares as

$$\sigma_j(\delta_t, x_t, \beta_d, \beta_\nu, J_t) = \int \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{j'} \exp\left(\delta_{j't} + \mu_{ij't}\right)} dF_\mu\left(\mu_{it} \mid x_t, \beta_d, \beta_\nu, J_t\right)$$

# Estimating mixed logit

*Inversion*

**Recall**: For logit model, we could easily recover vector $\delta_t$ that rationalized observed market shares. Can we do the same in the mixed-logit case?

$$\sigma_j(\delta_t, x_t, \beta_d, \beta_\nu, J_t) = \int \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{j'} \exp\left(\delta_{j't} + \mu_{ij't}\right)} dF_\mu\left(\mu_{it} \mid x_t, \beta_d, \beta_\nu, J_t\right)$$

**Inversion**: Given $\beta_d, \beta_\nu, x_t$, and observed shares $s$, find unique vector $\delta$ such that

$$\sigma_j(\delta, x_t, \beta_d, \beta_\nu, J_t) = \tilde{s}_j$$

# The BLP estimator

Start by partitioning the parameters:

$$\theta_1 \equiv (\alpha_0, \beta_0)$$
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In the literature, these are often referred to (perhaps somewhat misleadingly) as the “linear” and “non-linear” parameters, respectively. The actual distinction:

- $\theta_1$ affects mean utility $\delta$;
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Note: We can write

$$s_{jt} = \sigma_j(\delta_t, x_t, \theta_2, J_t)$$

# The BLP estimator

*Conceptual overview*

Suppose we have some $\theta$ and want to evaluate whether it fits the data well.

1. Invert: For each market, find $\delta_t$ such that $\tilde{s}_{jt} = \sigma_j(\delta_t, x_t, \theta_2, J_t)$ for all $j$, $t$. Call this $\delta_t(\theta_2)$.

   $\rightarrow$ This step only involves $\theta_2$.

2. Then, $\xi_{jt}(\theta) = \delta_{jt}(\theta_2) - (x_{jt}\beta_0 - \alpha_0 p_{jt})$.

   $\rightarrow$ This step only involves $\theta_1$.

3. Evaluate moment conditions of the form

$$\mathbb{E}\left[\xi_{jt}(\theta) z_{jt}\right]$$

   where $z_{jt} \supset x_{jt}$ is a vector of instruments. (We'll discuss the choice of instruments later on.)

# The BLP estimator

*Mathematical statement*

The BLP estimator $\hat{\theta}$ is the solution to

$$\begin{aligned}
\min_{\theta} \quad & g(\xi(\theta))' \, \Omega \, g(\xi(\theta)) \\
\text{subject to} \quad & g(\xi(\theta)) = \frac{1}{N} \sum_{\forall j,t} \xi_{jt}(\theta) z_{jt}, \\
& \xi_{jt}(\theta) = \delta_{jt}(\theta_2) - x_{jt}\beta + \alpha p_{jt}, \\
& \log(\tilde{s}_{jt}) = \log\left(\sigma_j(\delta_t, x_t, \theta_2, J_t)\right), \\
& \sigma_j(\delta_t, x_t, \theta_2, J_t) = \int \frac{\exp[\delta_{jt}(\theta_2) + x_{jt}\tilde{\beta}]}{1 + \sum_k \exp[\delta_{kt}(\theta_2) + x_{kt}\tilde{\beta}]} f_{\tilde{\beta}}(\tilde{\beta}|\theta_2) \, d\tilde{\beta}.
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**Note**: One approach to estimating demand is to solve this constrained optimization problem, treating $\theta, \xi$ as the choice variables. This is the MPEC approach of Dube, Fox, and Su (ECMA 2013).

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- Finally, we can regress $\delta_{jt}$ on observables using an appropriate **instrument** $z_{jt}$ for $p_{jt}$.

# Identification

*Random utility model*

An argument for nonparametric identification.

Suppose there is a characteristic $x_{jt}^{(1)}$ that enters utility linearly and doesn't have a random coefficient. Then, partition $x_{jt}$ as $(x_{jt}^{(1)}, x_{jt}^{(2)})$.

- **Index**: $\delta_{jt} = \beta^{(1)} x_{jt}^{(1)} + \xi_{jt}$. Utility depends on $x^{(1)}$ and $\xi$ only through $\delta$. (This is actually stronger than necessary.)
  - Scale normalization: Set $\beta^{(1)} = 1$. (You wouldn't do this in estimation. But in an identification argument, it's convenient—we don't have to carry around the $\beta^{(1)}$ parameter.)
- Condition on $x_t^{(2)} = x$. We can then think of identifying *separately for each* possible $x$.

# Identification

*Random utility model*

A key condition for inversion: (See Berry, Gandhi, and Haile (2013) and Berry and Haile (2014).)

— **Definition**: Let $\lambda_t$ denote either $-p_t$ or $\delta_t$. Goods $(0, 1, \ldots, J)$ are *connected substitutes in* $\lambda_t$ if both

(i) (Weak gross substitutes) $\sigma_k(\delta_t, p_t)$ is nonincreasing in $\lambda_{jt}$ for all $j > 0$, $k \neq j$, $(\delta_t, p_t) \in \mathbb{R}^{2J}$; and

(ii) (Some strict substitution) at each $(\delta_t, p_t) \in \text{supp}(\delta_t, p_t)$, for any nonempty $\mathcal{K} \subseteq \{1, \ldots, J\}$ there exists $k \in \mathcal{K}$ and $\ell \notin \mathcal{K}$ such that $\sigma_\ell(\delta_t, p_t)$ is strictly decreasing in $\lambda_{kt}$.

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- **Assumption**: Goods $(0, 1, \ldots, J)$ are connected substitutes in $-p_t$ and $\delta_t$.

*Note*: If (ii) failed, it would mean that there is some subset of goods that only substitute to other goods in the subset. This would mean that we've chosen our "market" definition poorly, i.e., $\{0, 1, \ldots, J\}$ is not really the relevant choice set.

# Identification

*Random utility model*

(Lemma 1, GH 2014) Consider any price vector $p$ and any market share vector $s$ such that $s_j > 0$ for all $j$ and $\sum_{j=1}^{J} s_j < 1$. Under the index assumption and the connected substitutes assumption, there is at most one vector $\delta$ such that

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- (Completeness, i.e., nonparametric version of rank) For all functions $B(s_t, p_t)$ with finite expectation, if $\mathbb{E}\left[B(s_t, p_t) \mid z_t, x_t^{(1)}\right] = 0$ a.s., then $B(s_t, p_t) = 0$ a.s.

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Let $B(s_t, p_t) = \sigma_j^{-1}(s_t, p_t) - \tilde{\sigma}_j^{-1}(s_t, p_t)$, we have

$$\mathbb{E}\left[B(s_t, p_t) \mid z_t, x_t^{(1)}\right] = 0, \quad \text{a.s.}$$

$$\Rightarrow \quad B(s_t, p_t) = 0, \quad \text{a.s.}$$

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*Random utility model*

**Theorem** (BH 2014): For all $j = 1, \ldots, J$, $\xi_{jt}$ is identified for all $t$ and $\sigma_j(\cdot)$ is identified.

**Proof sketch**: Suppose $\sigma_j$ and $\tilde{\sigma}_j$ both rationalize observed shares.

Recall that $\xi_{jt} = \sigma_j^{-1}(s_t, p_t) - x_{jt}^{(1)}$. Same is true for $\tilde{\sigma}_j$. Taking conditional expectations:

$$\mathbb{E}\left[\sigma_j^{-1}(s_t, p_t) \mid z_t, x_t^{(1)}\right] - x_{jt}^{(1)} = 0, \quad \text{a.s.}$$

$$\mathbb{E}\left[\tilde{\sigma}_j^{-1}(s_t, p_t) \mid z_t, x_t^{(1)}\right] - x_{jt}^{(1)} = 0, \quad \text{a.s.}$$

Let $B(s_t, p_t) = \sigma_j^{-1}(s_t, p_t) - \tilde{\sigma}_j^{-1}(s_t, p_t)$, we have

$$\mathbb{E}\left[B(s_t, p_t) \mid z_t, x_t^{(1)}\right] = 0, \quad \text{a.s.}$$

$$\Rightarrow \quad B(s_t, p_t) = 0, \quad \text{a.s.}$$

So, by completeness assumption, $\sigma_j^{-1} = \tilde{\sigma}_j^{-1}$ a.s.

# Outline

1. Challenges in demand estimation
2. Introduction to demand models
3. Overview of estimation
4. Identification
5. Instruments
6. Incorporating the supply side
7. Implementing the estimator
8. Applications
   - 8.1 Demand for cars (BLP)
   - 8.2 “Micro BLP”; welfare effects of introducing the minivan (Petrin)
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# Instruments

**Three common types of instruments**

1. Cost shifters and their proxies
   - Actual shifters: Exogenous material costs, tax, tariff, etc.
   - Noisy (cross-firm) shifters: Average wage level in market where produced
   - “Hausman instrument”: Average contemporaneous price of the same good in other markets.

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     - Markets are perfectly competitive; or
     - $\xi_{jt}$ and $\xi_{jt'}$ uncorrelated for all $t' \neq t$.
2. The "BLP instruments": Exogenous characteristics of competing products.
   - If $x_{jt}$ is mean-independent of $\xi_{jt}$, then it stands to reason that $x_{-jt}$ is too.
   - Shift $p_{jt}$ under many models of conduct.

# Instruments

**Three common types of instruments**

3. The “Waldfogel-Fan” instruments: Characteristics of nearby markets.
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   - *Example*: Average income in adjacent markets.

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3. The "Waldfogel-Fan" instruments: Characteristics of nearby markets.
   - Leverages the fact that firms often set prices at a region or "zone" level.
   - *Example*: Average income in adjacent markets.
   - Fan (2013): Two newspapers $j$, $k$ are sold in areas that partially overlap. Suppose the overlap is market $t$. Each sets a uniform price. Demographics within market $t'$ served *only* by $k$ affect $j$'s markups everywhere.

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# The supply side

*Why incorporate it?*

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*Why incorporate it?*

- Many counterfactuals of interest involve supply as well as demand. *Examples*:
  - Merger analysis
  - Analyzing new tax policy
- Even when we're solely interested in demand, supply model provides additional restrictions that can improve precision of estimates.
  - → Exploit exogeneity restrictions involving marginal cost shocks.

# Supply Side

- Marginal costs

$$c_{jt} = \bar{c}(Ms_{jt}, w_{jt}; \gamma) + \omega_{jt}$$

where $w_{jt}$ are cost shifters, $M$ is market size

  - Parametrize the marginal cost $\bar{c}$, e.g., as a linear function
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- Multiproduct firms price according to FOC:

$$p = c + \Delta^{-1}(p, x, \xi; \theta)s(p, x, \xi; \theta)$$

$$\text{where} \quad \Delta_{jr} = \begin{cases} -\partial s_{rt}/\partial p_{jt} & r \in \mathcal{F}_j \\ 0 & \text{otherwise} \end{cases}$$

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- Idea is to use the restrictions $E[(\xi, \omega)|z] = 0$ to create moments, where $z = (x, w)$
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  - **Question**: If we knew $\delta$ and $\theta, \gamma$, how would we compute $\xi, \omega$?
  - **Question**: For a given $\theta$, how would we get $\delta$?
- Put the pieces together:
  1. Pick demand parameters $\theta$ and cost parameters $\gamma$
     - Invert for $\delta_t = S^{-1}(s_t; x_t, p_t, \theta)$, substituting $\hat{s}$ for $s$ (may involve simulation—to be discussed below)
     - Calculate $\xi_{jt} = \delta_{jt} - x_{jt}\beta + \alpha p_{jt}$
     - Calculate $\omega_{jt} = p_{jt} - \bar{c}(Ms_{jt}, w_{jt}, \gamma) - \Delta_j^{-1}(p, x, \xi; \theta)s$
  2. Construct the GMM estimator $(\hat{\theta}, \hat{\gamma})$ $\arg\min \|G_J(\theta, \gamma)\|$ where

$$G_J(\theta, \gamma) = \frac{1}{\sum_t J_t} \sum (\xi_{jt}(\theta), \omega_{jt}(\theta, \gamma))' H_{jt}(z_t)(\xi_{jt}(\theta), \omega_{jt}(\theta, \gamma))$$

# Estimation

*Logit*

- Use the restrictions $E[(\xi, \omega)|z] = 0$, where $z = (x, w)$
- **Goal**: Find parameter values that minimize the GMM objective function: $(\hat{\theta}, \hat{\gamma}) = \arg\min \| G_J(\theta, \gamma) \|$

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$$\delta_{jt} = \log s_{jt} - \log s_{0t}$$

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$$\delta_{jt} = \log s_{jt} - \log s_{0t}$$

Then, for each $(\theta, \gamma)$ we consider, we compute the GMM objective:

1. Compute residuals:
   - $\xi_{jt} = \delta_{jt} - x_{jt}\beta + \alpha p_{jt}$
   - $\omega_{jt} = p_{jt} - \bar{c}(Ms_{jt}, w_{jt}, \gamma) - \Delta_j^{-1}(p, x, \xi; \theta)s$

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2. Construct the GMM objective

$$G_J(\theta, \gamma) = \frac{1}{\sum_t J_t} \sum (\xi_{jt}(\theta), \omega_{jt}(\theta, \gamma)) H_{jt}(z_t)$$

for some chosen functions $H_{\cdot}(z)$

# Estimation

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- Estimation
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     **Critical to note**: In the logit case, inversion $\delta_t = S^{-1}(s_t; x_t, p_t, \theta) = S^{-1}(s_t)$ does not depend on parameter values!

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     for some chosen functions $H_{\cdot}(z)$

# Estimation when inversion must be done repeatedly

*E.g., mixed logit*

- Use the restrictions $E[(\xi, \omega)|z] = 0$, where $z = (x, w)$
- **Goal**: Find parameter values that minimize the GMM objective function: $(\hat{\theta}, \hat{\gamma}) = \arg\min \|G_J(\theta, \gamma)\|$
- Estimation (more complex models, e.g., mixed logit)

For each $(\theta, \gamma)$ we consider, we compute the GMM objective:

1. Invert for $\delta_t = S^{-1}(s_t; x_t, p_t, \theta)$
2. Compute residuals:
   - $\xi_{jt} = \delta_{jt} - x_{jt}\beta + \alpha p_{jt}$
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for some chosen functions $H_{jt}(z)$.

# The BLP algorithm

GMM estimation: $\hat{\theta} = \text{argmin}_\theta \, G(\theta, \hat{s}, P)$

To evaluate $G(\theta, \hat{s}, P)$ for any given $\theta$,

1. Get $\xi(\theta, \hat{s}, P)$.
   - Use iteration to solve *fixed-point problem*.
     - For each update of iterative process, need to compute market share via *simulation*.
2. Get $\omega(\theta, \hat{s}, P)$.
   - Use price-markup equation to map prices to $\omega$.
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# Innermost steps: Computing shares and demand elasticities

Recall:

$$
\begin{aligned}
u_{ij} &= \overbrace{x_j\beta - \alpha p_j + \xi_j}^{\delta(x_j,p_j,\xi_j,\theta_1)} + \overbrace{\sum_k \sigma_k x_{jk}\nu_{ik}}^{\mu(x_j,p_j,\nu_i,\theta_2)} + \varepsilon_{ij} \\
f_j(\nu_i, \delta, p, x, \theta) &= \frac{\exp(\delta_j + \mu_{ij})}{1 + \sum_k \exp(\delta_k + \mu_{ik})} \\
s_j(p, x, \xi, \theta, P_0) &= \int f_j(\nu_i, \delta, p, x, \theta) P_0(d\nu)
\end{aligned}
$$

**Notes**:

- Market share conditional on $\nu$ (integrating over $\epsilon$) has closed form
- Aggregate market share (integrating over $\nu$) does not

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**Notes**:

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- So computing $s_j$ requires *simulation* (or maybe quadrature)

# Innermost steps: Computing shares and demand elasticities

$$
\begin{aligned}
s_j(p, x, \xi, \theta, P) &= \int f_j(\nu, \delta, p, x, \theta) P(d\nu) \\
&\approx \frac{1}{ns} \sum_{i=1}^{ns} f_j(\nu_i, \delta, p, x, \theta)
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where $(\nu_1, \ldots, \nu_{ns})$ are pseudo-random draws from $P$.

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And similarly for elasticities: **Questions**: Why do we need these elasticities? And how to compute them using what we know about logit demand?

$$\frac{\partial s_j}{\partial p_j} =$$
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# Innermost steps: Computing shares and demand elasticities

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$$\frac{\partial s_j}{\partial p_j} = \int f_j(\nu, \delta, p, x, \theta) \left(1 - f_j(\nu, \delta, p, x, \theta)\right) \left[\partial u_{ij} / \partial p_j\right] P(d_\nu)$$
$$\frac{\partial s_j}{\partial p_q} = -\int f_j(\nu, \delta, p, x, \theta) f_q(\nu, \delta, p, x, \theta) \left[\partial u_{ij} / \partial p_q\right] P(d_\nu)$$

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# Inversion: Find $\delta$ that rationalizes observed shares

System of $J$ (non-linear) equations in $J$ unknowns $(\delta)$.

$$\begin{bmatrix} \hat{s}_1 \\ \vdots \\ \hat{s}_J \end{bmatrix} = \begin{bmatrix} \int \frac{\exp(\delta_1+\mu_{i1})}{1+\sum_k \exp(\delta_k+\mu_{ik})} dF_\mu(\theta_2) \\ \vdots \\ \int \frac{\exp(\delta_J+\mu_{iJ})}{1+\sum_k \exp(\delta_k+\mu_{ik})} dF_\mu(\theta_2) \end{bmatrix}$$

More compactly,

$$\hat{s} = s(p, x, \delta, \theta_2, P_0)$$

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Equivalently,

$$\begin{aligned} \delta &= \delta + \log \hat{s} - \log s(p, x, \delta, \theta_2, P_0) \\ &\equiv T(s, \theta_2, P)[\delta] \end{aligned}$$

which can be shown to be a contraction mapping.

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which can be shown to be a contraction mapping.
**Question**: Why is contraction mapping important?

- $\rightarrow$ Guarantees unique solution.
- $\rightarrow$ Gives an iterative algorithm for solving the system: $\delta^{k+1} = T(s, \theta_2, P)[\delta^k]$.

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More compactly,

$$\hat{s} = s(p, x, \delta, \theta_2, P_0)$$

Equivalently,

$$\begin{aligned} \delta &= \delta + \log \hat{s} - \log s(p, x, \delta, \theta_2, P_0) \\ &\equiv T(s, \theta_2, P)[\delta] \end{aligned}$$

which can be shown to be a contraction mapping.

**Question**: Why is contraction mapping important?

→ Guarantees unique solution.

→ Gives an iterative algorithm for solving the system: $\delta^{k+1} = T(s, \theta_2, P)[\delta^k]$.

**Practical notes**: (1) Do this separately for each market. (2) If simulating, don't redraw $\nu$ every iteration.

Then, $\xi_j = \delta_j - x_j\beta + \alpha p_j$.

# Inversion: Find $\delta$ that rationalizes observed shares

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Then, $\xi_j = \delta_j - x_j\beta + \alpha p_j$. **Question**: Do you see an opportunity for improved computational efficiency related to this equation?

# The BLP algorithm

GMM estimation: $\hat{\theta} = \text{argmin}_\theta \, G(\theta, \hat{s}, P)$

To evaluate $G(\theta, \hat{s}, P)$ for any given $\theta$,

1. Get $\xi(\theta, \hat{s}, P)$.
   - Use iteration to solve *fixed-point problem*.
     - » For each update of iterative process, need to compute market share via *simulation*.
2. Get $\omega(\theta, \hat{s}, P)$.
   - **Use price-markup equation to map prices to $\omega$.**
     - » To compute markup, need to compute demand elasticities via *simulation*.
3. Compute moment function.

# Step 2: Find $\omega$ that rationalizes observed prices

From supply model and price-setting Nash equilibrium, (here, assuming a log-linear marginal cost function)

$$\ln(p - \Delta^{-1}(p, x, \xi; \theta)s(p, x, \xi; \theta)) = w\gamma + \omega$$

where

$$\Delta_{jr} = \begin{cases} -\partial s_{rt}/\partial p_{jt} & r \in \mathcal{F}_j \\ 0 & \text{otherwise} \end{cases}$$

# Issues to be aware of

- Practical/Empirical Issues:
  - Typically difficult to introduce many random coefficients—requires a fair amount of variation in characteristics across markets (i.e., years)
  - Market size must be picked (e.g., all US households). No perfect solution.

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- A lot of work on improving computation since BLP
  - Alternatives include MPEC (Dube, Fox and Su, 2012). Tends to work better than BLP when there are more than, say, five product characteristics.

✓ A great reference is Conlon and Gortomaker (2019) – pyBLP

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# Data and Motivation

*Berry, Levinsohn and Pakes (1995)*

- Context:
  - **Berry (1994)**: The "Berry inversion"; a method for estimating certain classes of demand models (e.g., logit and the vertical differentiation model we discussed last time)
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  - Address price endogeneity
  - Computationally tractable
- Data:
  - Aggregate car sales by model during 1971-1990
  - Model characteristics such as weight, horsepower, AC, MPG/MP$ etc.
  - Distribution of household income from the CPS

# Demand Model

- Model demand for cars using mixed (random coefficients) logit model (slightly simplified from BLP)

$$\begin{aligned} u_{ij} &= \overbrace{x_j\beta - \alpha p_j + \xi_j}^{\delta(x_j,p_j,\xi_j,\theta_1)} + \overbrace{\sum_k \sigma_k x_{jk}\nu_{ik}}^{\mu(x_j,p_j,\nu_i,\theta_2)} + \varepsilon_{ij} \\ u_{i0} &= 0 \end{aligned}$$

  where $\nu_{ik}$ is iid across $ik$ with known distribution (e.g., standard Normal) and $\varepsilon_{ij}$ is iid across $ij$ w/ EV I

- Then,

$$s_j = \int \frac{\exp(\delta_j + \mu_{ij})}{1 + \sum_k \exp(\delta_k + \mu_{ik})} dF_\mu(\theta_2)$$

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    - » Yugo and Mercedes have same market share $\rightarrow$ logit restricts substitution. (**How?**)

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  - Consumers differ on price sensitivity due to income (BLP use $y_i = \nu_i$ distr. from CPS)

TABLE II

THE RANGE OF CONTINUOUS DEMAND CHARACTERISTICS
(AND ASSOCIATED MODELS)

| Variable | Percentile | | | | |
|---|---|---|---|---|---|
| | 0 | 25 | 50 | 75 | 100 |
| *Price* | 90 Yugo | 79 Mercury Capri | 87 Buick Skylark | 71 Ford T-Bird | 89 Porsche 911 Cabriolet |
| | 3.393 | 6.711 | 8.728 | 13.074 | 68.597 |
| *Sales* | 73 Toyota 1600CR | 72 Porsche Rdstr | 77 Plym. Arrow | 82 Buick LeSabre | 71 Chevy Impala |
| | .049 | 15.479 | 47.345 | 109.002 | 577.313 |
| *HP/Wt.* | 85 Plym. Gran Fury | 85 Suburu DH | 86 Plym. Caravelle | 89 Toyota Camry | 89 Porsche 911 Turbo |
| | 0.170 | 0.337 | 0.375 | 0.428 | 0.948 |
| *Size* | 73 Honda Civic | 77 Renault GTL | 89 Hyundai Sonata | 81 Pontiac F-Bird | 73 Imperial |
| | 0.756 | 1.131 | 1.270 | 1.453 | 1.888 |
| *MP$* | 74 Cad. Eldorado | 78 Buick Skyhawk | 82 Mazda 626 | 84 Pontiac 2000 | 89 Geo Metro |
| | 8.46 | 15.57 | 20.10 | 24.86 | 64.37 |
| *MPG* | 74 Cad. Eldorado | 79 BMW 528i | 81 Dodge Challenger | 75 Suburu DL | 89 Geo Metro |
| | 9 | 17 | 20 | 25 | 53 |

TABLE III

RESULTS WITH LOGIT DEMAND AND MARGINAL COST PRICING
(2217 OBSERVATIONS)

| Variable | OLS Logit Demand | IV Logit Demand | OLS ln (*price*) on *w* |
|---|---|---|---|
| Constant | −10.068 | −9.273 | 1.882 |
| | (0.253) | (0.493) | (0.119) |
| *HP/Weight** | −0.121 | 1.965 | 0.520 |
| | (0.277) | (0.909) | (0.035) |
| *Air* | −0.035 | 1.289 | 0.680 |
| | (0.073) | (0.248) | (0.019) |
| *MP$* | 0.263 | 0.052 | — |
| | (0.043) | (0.086) | |
| *MPG** | — | — | −0.471 |
| | | | (0.049) |
| *Size** | 2.341 | 2.355 | 0.125 |
| | (0.125) | (0.247) | (0.063) |
| *Trend* | — | — | 0.013 |
| | | | (0.002) |
| *Price* | −0.089 | −0.216 | — |
| | (0.004) | (0.123) | |
| *No. Inelastic Demands* (+/−2 *s.e.'s*) | 1494 (1429–1617) | 22 (7–101) | *n.a.* |
| $R^2$ | 0.387 | *n.a.* | .656 |

- **Question**: Why do instruments increase magnitude of price coefficient?
- Without instruments, many firms price on inelastic portion of the demand curve.

TABLE IV

ESTIMATED PARAMETERS OF THE DEMAND AND PRICING EQUATIONS: BLP SPECIFICATION, 2217 OBSERVATIONS

| Demand Side Parameters | Variable | Parameter Estimate | Standard Error | Parameter Estimate | Standard Error |
|---|---|---|---|---|---|
| Means ($\bar{\beta}$'s) | *Constant* | −7.061 | 0.941 | −7.304 | 0.746 |
| | *HP/Weight* | 2.883 | 2.019 | 2.185 | 0.896 |
| | *Air* | 1.521 | 0.891 | 0.579 | 0.632 |
| | *MP$* | −0.122 | 0.320 | −0.049 | 0.164 |
| | *Size* | 3.460 | 0.610 | 2.604 | 0.285 |
| Std. Deviations ($\sigma_\beta$'s) | *Constant* | 3.612 | 1.485 | 2.009 | 1.017 |
| | *HP/Weight* | 4.628 | 1.885 | 1.586 | 1.186 |
| | *Air* | 1.818 | 1.695 | 1.215 | 1.149 |
| | *MP$* | 1.050 | 0.272 | 0.670 | 0.168 |
| | *Size* | 2.056 | 0.585 | 1.510 | 0.297 |
| Term on Price ($\alpha$) | $\ln(y-p)$ | 43.501 | 6.427 | 23.710 | 4.079 |
| Cost Side Parameters | | | | | |
| | *Constant* | 0.952 | 0.194 | 0.726 | 0.285 |
| | $\ln(HP/Weight)$ | 0.477 | 0.056 | 0.313 | 0.071 |
| | *Air* | 0.619 | 0.038 | 0.290 | 0.052 |
| | $\ln(MPG)$ | −0.415 | 0.055 | 0.293 | 0.091 |
| | $\ln(Size)$ | −0.046 | 0.081 | 1.499 | 0.139 |
| | *Trend* | 0.019 | 0.002 | 0.026 | 0.004 |
| | $\ln(q)$ | | | −0.387 | 0.029 |

- Appears to be returns to scale, but probably too large to be plausible
- **Question**: How does high $\bar{\beta}_k$ differ from high $\sigma_{\beta,k}$ in terms of implications for substitution patterns?

TABLE VI

A SAMPLE FROM 1990 OF ESTIMATED OWN- AND CROSS-PRICE SE
BASED ON TABLE IV (CRTS) ESTIMATES

| | Mazda 323 | Nissan Sentra | Ford Escort | Chevy Cavalier | Honda Accord | Ford Taurus | Buick Century | Nissan Maxima |
|---|---|---|---|---|---|---|---|---|
| 323 | −125.933 | 1.518 | 8.954 | 9.680 | 2.185 | 0.852 | 0.485 | 0.056 |
| Sentra | 0.705 | −115.319 | 8.024 | 8.435 | 2.473 | 0.909 | 0.516 | 0.093 |
| Escort | 0.713 | 1.375 | −106.497 | 7.570 | 2.298 | 0.708 | 0.445 | 0.082 |
| Cavalier | 0.754 | 1.414 | 7.406 | −110.972 | 2.291 | 1.083 | 0.646 | 0.087 |
| Accord | 0.120 | 0.293 | 1.590 | 1.621 | −51.637 | 1.532 | 0.463 | 0.310 |
| Taurus | 0.063 | 0.144 | 0.653 | 1.020 | 2.041 | −43.634 | 0.335 | 0.245 |
| Century | 0.099 | 0.228 | 1.146 | 1.700 | 1.722 | 0.937 | −66.635 | 0.773 |
| Maxima | 0.013 | 0.046 | 0.236 | 0.256 | 1.293 | 0.768 | 0.866 | −35.378 |
| Legend | 0.004 | 0.014 | 0.083 | 0.084 | 0.736 | 0.532 | 0.318 | 0.506 |
| TownCar | 0.002 | 0.006 | 0.029 | 0.046 | 0.475 | 0.614 | 0.210 | 0.389 |
| Seville | 0.001 | 0.005 | 0.026 | 0.035 | 0.425 | 0.420 | 0.131 | 0.351 |
| LS400 | 0.001 | 0.003 | 0.018 | 0.019 | 0.302 | 0.185 | 0.079 | 0.280 |
| 735*i* | 0.000 | 0.002 | 0.009 | 0.012 | 0.203 | 0.176 | 0.050 | 0.190 |

- Rich cross-price elasticities

# Substitution to the outside good

**Question**: In the logit model, what is diversion ratio between good $j$ and the outside good?

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TABLE VII

SUBSTITUTION TO THE OUTSIDE GOOD

| Model | Given a price increase, the percentage who substitute to the outside good (as a percentage of all who substitute away.) Logit | BLP |
|---|---|---|
| Mazda 323 | 90.870 | 27.123 |
| Nissan Sentra | 90.843 | 26.133 |
| Ford Escort | 90.592 | 27.996 |
| Chevy Cavalier | 90.585 | 26.389 |
| Honda Accord | 90.458 | 21.839 |
| Ford Taurus | 90.566 | 25.214 |
| Buick Century | 90.777 | 25.402 |
| Nissan Maxima | 90.790 | 21.738 |
| Acura Legend | 90.838 | 20.786 |
| Lincoln Town Car | 90.739 | 20.309 |
| Cadillac Seville | 90.860 | 16.734 |
| Lexus LS400 | 90.851 | 10.090 |
| BMW 735*i* | 90.883 | 10.101 |

# Markups and profits

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TABLE VIII

A SAMPLE FROM 1990 OF ESTIMATED PRICE-MARGINAL COST MARKUPS AND VARIABLE PROFITS: BASED ON TABLE 6 (CRTS) ESTIMATES

| | Price | Markup Over MC $(p - MC)$ | Variable Profits (in \$'000's) $q*(p - MC)$ |
|---|---|---|---|
| Mazda 323 | \$5,049 | \$ 801 | \$18,407 |
| Nissan Sentra | \$5,661 | \$ 880 | \$43,554 |
| Ford Escort | \$5,663 | \$1,077 | \$311,068 |
| Chevy Cavalier | \$5,797 | \$1,302 | \$384,263 |
| Honda Accord | \$9,292 | \$1,992 | \$830,842 |
| Ford Taurus | \$9,671 | \$2,577 | \$807,212 |
| Buick Century | \$10,138 | \$2,420 | \$271,446 |
| Nissan Maxima | \$13,695 | \$2,881 | \$288,291 |
| Acura Legend | \$18,944 | \$4,671 | \$250,695 |
| Lincoln Town Car | \$21,412 | \$5,596 | \$832,082 |
| Cadillac Seville | \$24,353 | \$7,500 | \$249,195 |
| Lexus LS400 | \$27,544 | \$9,030 | \$371,123 |
| BMW 735*i* | \$37,490 | \$10,975 | \$114,802 |

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- In fact, they are sometimes imprecise in dimensions that are particularly important for our counterfactuals of interest.
  - Income effects: BLP estimates imply that only the wealthiest households purchase new vehicles.
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  - Income effects: BLP estimates imply that only the wealthiest households purchase new vehicles.
  - Idiosyncratic $\epsilon$ plays a big role in rationalizing choices.
- **Question**: Suppose we want to evaluate the welfare effects of the introduction of minivans in 1984. Why would the above pose a problem?

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- **Key idea**: We should exploit this additional information to improve demand estimates.
- We do this by adding an additional set of moments; the idea is that the model should match features of the micro data. Petrin's micro-moments:
  - Purchase probability and income: $\Pr(i \text{ buys new vehicle} \mid \bar{y}_1 \leqslant y_i < \bar{y}_2)$.
  - Family size and vehicle type:

$$\mathbb{E}\left[\text{family size}_i \mid i \text{ purchases a minivan}\right]$$
$$\mathbb{E}\left[\text{family size}_i \mid i \text{ purchases a station wagon}\right]$$
$$\vdots$$

# Micro BLP (Petrin 2002)

Call these moments $f_m(\bar{v})$ for $m = 1, \ldots, M$. Call the model analog $f_m(v(\theta))$. Then,

$$g(\theta) = \begin{bmatrix} g_{\text{Agg}}(\theta) \\ g_{\text{Micro}}(\theta) \end{bmatrix}$$

where

$$g_{\text{Micro}}(\theta) = \begin{bmatrix} f_1(\bar{v}) - f_1(v(\theta)) \\ \vdots \\ f_M(\bar{v}) - f_M(v(\theta)) \end{bmatrix}$$

and $g_{\text{Agg}}(\theta)$ are the aggregate moments we had before.

# Micro BLP (Petrin 2002)

TABLE 4
PARAMETER ESTIMATES FOR THE DEMAND-SIDE EQUATION

| Variable | OLS Logit (1) | Instrumental Variable Logit (2) | (BLP) Random Coefficients (3) | Random Coefficients and Microdata (4) |
|---|---|---|---|---|
| | | A. Price Coefficients ($\alpha$'s) | | |
| $\alpha_1$ | .07 | .13 | 4.92 | |
| | (.01)** | (.01)** | (9.78) | |
| $\alpha_2$ | | | 11.89 | |
| | | | (21.41) | |
| $\alpha_3$ | | | 37.92 | |
| | | | (18.64)** | |
| | | B. Base Coefficients ($\beta$'s) | | |
| Constant | −10.03 | −10.04 | −12.74 | |
| | (.32)** | (.34)** | (5.65)** | |
| Horsepower/weight | 1.48 | 3.78 | 3.40 | |
| | (.34)** | (.44)** | (39.79) | |
| Size | 3.17 | 3.25 | 4.60 | |
| | (.26)** | (.27)** | (24.64) | |
| Air conditioning standard | −.20 | .21 | −1.97 | |
| | (.06)** | (.08)** | (2.23) | |
| Miles/dollar | .18 | .05 | −.54 | |
| | (.06)** | (.07) | (3.40) | |
| Front wheel drive | .32 | .15 | −5.24 | |
| | (.05)** | (.06)** | (3.09) | |
| Minivan | .09 | −.10 | −4.34 | |
| | (.14) | (.15) | (13.16) | |
| Station wagon | −1.12 | −1.12 | −20.52 | |
| | (.06)** | (.07)** | (36.17) | |
| Sport-utility | −.41 | −.61 | −3.10 | |
| | (.09)** | (.10)** | (10.76) | |
| Full-size van | −1.73 | −1.89 | −28.54 | |
| | (.16)** | (.17)** | (235.51) | |
| % change GNP | .03 | .03 | .08 | |
| | (.01)** | (.01)** | (.02)** | |

NOTE.—Standard errors are in parentheses. A quadratic time trend is included in all specifications.
* Zstatistic >1.
** Zstatistic >2.

# Micro BLP (Petrin 2002)

TABLE 4
PARAMETER ESTIMATES FOR THE DEMAND-SIDE EQUATION

| Variable | OLS Logit (1) | Instrumental Variable Logit (2) | (BLP) Random Coefficients (3) | Random Coefficients and Microdata (4) |
|---|---|---|---|---|
| | A. Price Coefficients ($\alpha$'s) | | | |
| $\alpha_1$ | .07 | .13 | 4.92 | 7.52 |
| | (.01)** | (.01)** | (9.78) | (1.24)** |
| $\alpha_2$ | | | 11.89 | 31.13 |
| | | | (21.41) | (4.07)** |
| $\alpha_3$ | | | 37.92 | 34.49 |
| | | | (18.64)** | (2.56)** |
| | B. Base Coefficients ($\beta$'s) | | | |
| Constant | −10.03 | −10.04 | −12.74 | −15.67 |
| | (.32)** | (.34)** | (5.65)** | (4.39)** |
| Horsepower/weight | 1.48 | 3.78 | 3.40 | −2.83 |
| | (.34)** | (.44)** | (39.79) | (8.16) |
| Size | 3.17 | 3.25 | 4.60 | 4.80 |
| | (.26)** | (.27)** | (24.64) | (3.57)* |
| Air conditioning standard | −.20 | .21 | −1.97 | 3.88 |
| | (.06)** | (.08)** | (2.23) | (2.21)* |
| Miles/dollar | .18 | .05 | −.54 | −15.79 |
| | (.06)** | (.07) | (3.40) | (.87)** |
| Front wheel drive | .32 | .15 | −5.24 | −12.32 |
| | (.05)** | (.06)** | (3.09) | (2.36)** |
| Minivan | .09 | −.10 | −4.34 | −5.65 |
| | (.14) | (.15) | (13.16) | (.68)** |
| Station wagon | −1.12 | −1.12 | −20.52 | −1.31 |
| | (.06)** | (.07)** | (36.17) | (.36)** |
| Sport-utility | −.41 | −.61 | −3.10 | −4.38 |
| | (.09)** | (.10)** | (10.76) | (.41)** |
| Full-size van | −1.73 | −1.89 | −28.54 | −5.26 |
| | (.16)** | (.17)** | (235.51) | (1.30)** |
| % change GNP | .03 | .03 | .08 | .24 |
| | (.01)** | (.01)** | (.02)** | (.02)** |

NOTE.—Standard errors are in parentheses. A quadratic time trend is included in all specifications.
* Zstatistic >1.
** Zstatistic >2.

# Micro BLP (Petrin 2002)

TABLE 8
AVERAGE COMPENSATING VARIATION CONDITIONAL ON MINIVAN PURCHASE, 1984: 1982–84 CPI-ADJUSTED DOLLARS

| | OLS Logit | Instrumental Variable Logit | Random Coefficients | Random Coefficients and Microdata |
|---|---|---|---|---|
| Compensating variation: | | | | |
| Median | 9,573 | 5,130 | | |
| Mean | 13,652 | 7,414 | | |
| Welfare change from difference in: | | | | |
| Observed characteristics ($\delta_j+\mu_{ij}$) | −81,469 | −44,249 | | |
| Logit Error ($\epsilon_{ij}$) | 95,121 | 51,663 | | |
| Income of minivan purchasers: | | | | |
| Estimate from model | 23,728 | 23,728 | | |
| Difference from actual (CEX) | −15,748 | −15,748 | | |

NOTE.—Compensating variation is evaluated at equilibrium prices without minivans. Decomposition of compensation is the average difference in the value of observed and unobserved characteristics between first and second choices. For logit models, the purchase decision is independent of income, so mean purchaser income is mean U.S. household income.

# Micro BLP (Petrin 2002)

TABLE 8
AVERAGE COMPENSATING VARIATION CONDITIONAL ON MINIVAN PURCHASE, 1984: 1982–84 CPI-ADJUSTED DOLLARS

| | OLS Logit | Instrumental Variable Logit | Random Coefficients | Random Coefficients and Microdata |
|---|---|---|---|---|
| Compensating variation: | | | | |
| Median | 9,573 | 5,130 | 1,217 | |
| Mean | 13,652 | 7,414 | 3,171 | |
| Welfare change from difference in: | | | | |
| Observed characteristics ($\delta_j+\mu_{ij}$) | −81,469 | −44,249 | −820 | |
| Logit Error ($\epsilon_{ij}$) | 95,121 | 51,663 | 3,991 | |
| Income of minivan purchasers: | | | | |
| Estimate from model | 23,728 | 23,728 | 99,018 | |
| Difference from actual (CEX) | −15,748 | −15,748 | 59,542 | |

NOTE.—Compensating variation is evaluated at equilibrium prices without minivans. Decomposition of compensation is the average difference in the value of observed and unobserved characteristics between first and second choices. For logit models, the purchase decision is independent of income, so mean purchaser income is mean U.S. household income.

# Micro BLP (Petrin 2002)

TABLE 8
AVERAGE COMPENSATING VARIATION CONDITIONAL ON MINIVAN PURCHASE, 1984: 1982–84 CPI-ADJUSTED DOLLARS

| | OLS Logit | Instrumental Variable Logit | Random Coefficients | Random Coefficients and Microdata |
|---|---|---|---|---|
| Compensating variation: | | | | |
| Median | 9,573 | 5,130 | 1,217 | 783 |
| Mean | 13,652 | 7,414 | 3,171 | 1,247 |
| Welfare change from difference in: | | | | |
| Observed characteristics ($\delta_j+\mu_{ij}$) | −81,469 | −44,249 | −820 | 851 |
| Logit Error ($\epsilon_{ij}$) | 95,121 | 51,663 | 3,991 | 396 |
| Income of minivan purchasers: | | | | |
| Estimate from model | 23,728 | 23,728 | 99,018 | 36,091 |
| Difference from actual (CEX) | −15,748 | −15,748 | 59,542 | −3,385 |

NOTE.—Compensating variation is evaluated at equilibrium prices without minivans. Decomposition of compensation is the average difference in the value of observed and unobserved characteristics between first and second choices. For logit models, the purchase decision is independent of income, so mean purchaser income is mean U.S. household income.

# Outline

1. Challenges in demand estimation
2. Introduction to demand models
3. Overview of estimation
4. Identification
5. Instruments
6. Incorporating the supply side
7. Implementing the estimator
8. Applications
   - 8.1 Demand for cars (BLP)
   - 8.2 “Micro BLP”; welfare effects of introducing the minivan (Petrin)
   - 8.3 Market power in the ready-to-eat cereal industry (Nevo)

# Nevo (2001): Market Power in the Cereal Industry

# Market Power in the Cereal Industry

*Motivation and question*

# Market Power in the Cereal Industry

*Motivation and question*

- Motivating facts—RTE cereal looks like an industry with lots of market power:
  - High concentration, large (accounting) margins, lots of advertising, lots of introduction of new products
- The paper's main question: Are the high price-cost margins in the RTE cereal industry a result of collusion?

# Market Power in the Cereal Industry

*Motivation and question*

- Motivating facts—RTE cereal looks like an industry with lots of market power:
  - High concentration, large (accounting) margins, lots of advertising, lots of introduction of new products
- The paper's main question: Are the high price-cost margins in the RTE cereal industry a result of collusion?
- **Question**: At a high level, what is the approach Nevo takes to answering this question?

# Market Power in the Cereal Industry

## *Approach*

Decompose margins into

- Product differentiation
- Multi-product pricing effects
- Potential collusion

Approach

- Use only *demand estimates* to recover markups under alternative models of firm conduct
- Panel data on demand for cereals across geographical markets

# Market Power in the Cereal Industry

*Approach—a bit more detail*

1. Estimate a model of demand.
2. Use demand system to compute markups under three hypothetical industry structures:
   - (i) Single-product firms
   - (ii) Multi-product firms (as given by the actual brand ownership of products)
   - (iii) Joint monopoly (i.e., full collusion)
3. Check which of these gives markups closest to those obtained from crude accounting measure

# Motivating Facts

*A highly concentrated industry*

TABLE I

VOLUME MARKET SHARES

| | 88Q1 | 88Q4 | 89Q4 | 90Q4 | 91Q4 | 92Q4 |
|---|---|---|---|---|---|---|
| Kellogg | 41.39 | 39.91 | 38.49 | 37.86 | 37.48 | 33.70 |
| General Mills | 22.04 | 22.30 | 23.60 | 23.82 | 25.33 | 26.83 |
| Post | 11.80 | 10.30 | 9.45 | 10.96 | 11.37 | 11.31 |
| Quaker Oats | 9.93 | 9.00 | 8.29 | 7.66 | 7.00 | 7.40 |
| Ralston | 4.86 | 6.37 | 7.65 | 6.60 | 5.45 | 5.18 |
| Nabisco | 5.32 | 6.01 | 4.46 | 3.75 | 2.95 | 3.11 |
| C3 | 75.23 | 72.51 | 71.54 | 72.64 | 74.18 | 71.84 |
| C6 | 95.34 | 93.89 | 91.94 | 90.65 | 89.58 | 87.53 |
| Private Label | 3.33 | 3.75 | 4.63 | 6.29 | 7.13 | 7.60 |

# Motivating Facts

*Large gross (accounting) markups*

TABLE II

AGGREGATE ESTIMATES OF PRODUCTION COSTS

| Item | RTE Cereal (SIC 2043) | | All Food Industries (SIC 20) | |
|---|---|---|---|---|
| | M$ | % of value | M$ | % of value |
| Value of Shipments | 8,211 | 100.0 | 371,246 | 100.0 |
| Materials | 2,179 | 26.5 | 235,306 | 63.4 |
| Labor | 677 | 8.2 | 32,840 | 8.8 |
| Energy | 76 | 0.9 | 4,882 | 1.3 |
| Gross Margin | | 64.4 | | 26.5 |

*Source:* Annual Survey of Manufacturers 1988–1992.

# Supply Side

- Markup given by

$$p - mc = \Delta^{-1} s(p)$$

where various choices of

$$\Delta_{jr} = \begin{cases} -\partial s_{rt}/\partial p_{jt} & j, r \text{ price jointly} \\ 0 & \text{otherwise} \end{cases}$$

control whether

1. Single product pricing is occuring
2. Multi-product pricing
3. Collusive behavior

# Decomposition of markups

| | Product diff. | Portfolio effect | Collusion |
|---|---|---|---|
| Single-product | X | | |
| Multi-product | X | X | |
| Joint monopoly | X | X | X |

# Demand and Instruments

- Demand model is similar to BLP

$$u_{ijt} = x_j \beta_i^* - \alpha_i^* p_{jt} + \xi_j + \Delta\xi_{jt} + \varepsilon_{ijt}$$

where

$$\begin{pmatrix} \alpha_i^* \\ \beta_i^* \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \Pi D_i + \Sigma \nu_i$$

with $\nu_i \sim N(0, I_{k+1})$ and demographics $D_i$. $t$ is city-quarter (6 quarters)

# Demand and Instruments

- Demand model is similar to BLP

$$u_{ijt} = x_j \beta_i^* - \alpha_i^* p_{jt} + \xi_j + \Delta\xi_{jt} + \varepsilon_{ijt}$$

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- $\xi_j$ is a brand dummy
  - Wish to identify with city-quarter-brand variation in prices and $\Delta\xi_{jt}$
  - BLP instruments are constant within-brand (constant characteristics)

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  - BLP instruments are constant within-brand (constant characteristics)

- Two instruments
  1. Hausman (1996) style instruments – prices in other cities
     ✓ Idea: mc is common, demand shocks are idiosyncratic.
     Problems: Common national shocks, coordinated advertising/stocking
  2. "Cost-side" instruments: region dummies to pick up transportation costs; city density (cost of space); supermarket earnings from CPS
     Problems: Brand-specific regional shocks, or changes in demand due to earnings

# Demand and instruments

- Demand model is similar to BLP

$$u_{ijt} = x_j \beta_i^* - \alpha_i^* p_{jt} + \xi_j + \Delta\xi_{jt} + \varepsilon_{ijt}$$

where

$$\begin{pmatrix} \alpha_i^* \\ \beta_i^* \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \Pi D_i + \Sigma \nu_i$$

with $\nu_i \sim N(0, I_{k+1})$ and demographics $D_i$. $t$ is city-quarter (6 quarters)

- $\xi_j$ is a brand dummy
  - Wish to identify with city-quarter-brand variation in prices and $\Delta\xi_{jt}$
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- **Question**: Including the brand dummies seems like a big headache. So why bother?

# Demand and instruments

- Demand model is similar to BLP

$$u_{ijt} = x_j \beta_i^* - \alpha_i^* p_{jt} + \xi_j + \Delta\xi_{jt} + \varepsilon_{ijt}$$

where

$$\begin{pmatrix} \alpha_i^* \\ \beta_i^* \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \Pi D_i + \Sigma \nu_i$$

with $\nu_i \sim N(0, I_{k+1})$ and demographics $D_i$. $t$ is city-quarter (6 quarters)

- $\xi_j$ is a brand dummy
  - Wish to identify with city-quarter-brand variation in prices and $\Delta\xi_{jt}$
  - BLP instruments are constant within-brand (constant characteristics)
- **Question**: Including the brand dummies seems like a big headache. So why bother?

1. $x$ characteristics may not be a great description of products relevant characteristics.
2. Correlation between price and time- and market-invariant $\xi_j$.

# Price Variation

TABLE IV

PRICES AND MARKET SHARES OF BRANDS IN SAMPLE

| Description | Mean | Median | Std | Min | Max | Brand Variation | City Variation | Quarter Variation |
|---|---|---|---|---|---|---|---|---|
| Prices (¢ per serving) | 19.4 | 18.9 | 4.8 | 7.6 | 40.9 | 88.4% | 5.3% | 1.6% |
| Advertising (M$ per quarter) | 3.56 | 3.04 | 2.03 | 0 | 9.95 | 66.2% | — | 1.8% |
| Share within Cereal Market (%) | 2.2 | 1.6 | 1.6 | 0.1 | 11.6 | 82.3% | 0.5% | 0% |

*Source:* IRI Infoscan Data Base, University of Connecticut, Food Marketing Center.

TABLE V

RESULTS FROM LOGIT DEMAND[a]

| Variable | OLS | | | IV | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | (i) | (ii) | (iii) | (iv) | (v) | (vi) | (vii) | (viii) | (ix) | (x) |
| Price | −4.96 | −7.26 | −7.97 | −8.17 | −17.57 | −17.12 | −22.56 | −23.77 | −23.37 | −23.07 |
| | (0.10) | (0.16) | (0.15) | (0.11) | (0.50) | (0.49) | (0.51) | (0.53) | (0.47) | (1.17) |
| Advertising | 0.158 | 0.026 | 0.026 | 0.157 | 0.020 | 0.020 | 0.018 | 0.017 | 0.018 | 0.013 |
| | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) |
| Log of Median Income | — | — | 0.89 | — | — | — | 1.06 | 1.13 | 1.12 | — |
| | | | (0.02) | | | | (0.02) | (0.02) | (0.02) | |
| Log of Median Age | — | — | −0.423 | — | — | — | −0.063 | 0.003 | −0.007 | — |
| | | | (0.052) | | | | (0.059) | (0.062) | (0.061) | |
| Median HH Size | — | — | −0.126 | — | — | — | −0.053 | −0.036 | −0.038 | — |
| | | | (0.027) | | | | (0.029) | (0.031) | (0.031) | |
| Fit/Test of Over Identification[b] | 0.54 | 0.72 | 0.74 | 436.9 | 168.5 | 181.2 | 83.96 | 82.95 | 85.87 | 15.06 |
| | | | | (26.30) | (30.14) | (16.92) | (30.14) | (16.92) | (42.56) | (42.56) |
| 1st Stage $R^2$ | — | — | — | 0.889 | 0.908 | 0.908 | 0.910 | 0.909 | 0.913 | 0.952 |
| 1st Stage $F$-test | — | — | — | 5119 | 124 | 288 | 129 | 291 | 144 | 180 |
| Instruments[c] | — | — | — | brand dummies | prices | cost | prices | cost | prices, cost | prices, cost |

— **Question**: What purpose do these logit estimates serve?

TABLE V

RESULTS FROM LOGIT DEMAND[a]

| Variable | OLS | | | IV | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | (i) | (ii) | (iii) | (iv) | (v) | (vi) | (vii) | (viii) | (ix) | (x) |
| Price | −4.96 | −7.26 | −7.97 | −8.17 | −17.57 | −17.12 | −22.56 | −23.77 | −23.37 | −23.07 |
| | (0.10) | (0.16) | (0.15) | (0.11) | (0.50) | (0.49) | (0.51) | (0.53) | (0.47) | (1.17) |
| Advertising | 0.158 | 0.026 | 0.026 | 0.157 | 0.020 | 0.020 | 0.018 | 0.017 | 0.018 | 0.013 |
| | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) |
| Log of Median Income | — | — | 0.89 | — | — | — | 1.06 | 1.13 | 1.12 | — |
| | | | (0.02) | | | | (0.02) | (0.02) | (0.02) | |
| Log of Median Age | — | — | −0.423 | — | — | — | −0.063 | 0.003 | −0.007 | — |
| | | | (0.052) | | | | (0.059) | (0.062) | (0.061) | |
| Median HH Size | — | — | −0.126 | — | — | — | −0.053 | −0.036 | −0.038 | — |
| | | | (0.027) | | | | (0.029) | (0.031) | (0.031) | |
| Fit/Test of Over Identification[b] | 0.54 | 0.72 | 0.74 | 436.9 | 168.5 | 181.2 | 83.96 | 82.95 | 85.87 | 15.06 |
| | | | | (26.30) | (30.14) | (16.92) | (30.14) | (16.92) | (42.56) | (42.56) |
| 1st Stage $R^2$ | — | — | — | 0.889 | 0.908 | 0.908 | 0.910 | 0.909 | 0.913 | 0.952 |
| 1st Stage $F$-test | — | — | — | 5119 | 124 | 288 | 129 | 291 | 144 | 180 |
| Instruments[c] | — | — | — | brand dummies | prices | cost | prices | cost | prices, cost | prices, cost |

- **Question**: What purpose do these logit estimates serve?
- A paper that provides a framework for thinking about instrument choices in structural models: Andrews, Gentzkow, and Shapiro (2017)

TABLE VI

RESULTS FROM THE FULL MODEL[a]

| Variable | Means ($\beta$'s) | Standard Deviations ($\sigma$'s) | Interactions with Demographic Variables: Income | Income Sq | Age | Child |
|---|---|---|---|---|---|---|
| Price | −27.198 | 2.453 | 315.894 | −18.200 | — | 7.634 |
| | (5.248) | (2.978) | (110.385) | (5.914) | | (2.238) |
| Advertising | 0.020 | — | — | — | — | — |
| | (0.005) | | | | | |
| Constant | −3.592[b] | 0.330 | 5.482 | — | 0.204 | — |
| | (0.138) | (0.609) | (1.504) | | (0.341) | |
| Cal from Fat | 1.146[b] | 1.624 | — | — | — | — |
| | (0.128) | (2.809) | | | | |
| Sugar | 5.742[b] | 1.661 | −24.931 | — | 5.105 | — |
| | (0.581) | (5.866) | (9.167) | | (3.418) | |
| Mushy | −0.565[b] | 0.244 | 1.265 | — | 0.809 | — |
| | (0.052) | (0.623) | (0.737) | | (0.385) | |
| Fiber | 1.627[b] | 0.195 | — | — | — | −0.110 |
| | (0.263) | (3.541) | | | | (0.0513) |
| All-family | 0.781[b] | 0.1330 | — | — | — | |
| | (0.075) | (1.365) | | | | |
| Kids | 1.021[b] | 2.031 | — | — | — | |
| | (0.168) | (0.448) | | | | |
| Adults | 1.972[b] | 0.247 | — | — | — | |
| | (0.186) | (1.636) | | | | |
| GMM Objective (degrees of freedom) | | | 5.05 (8) | | | |
| MD $\chi^2$ | | | 3472.3 | | | |
| % of Price Coefficients > 0 | | | 0.7 | | | |

TABLE VII

MEDIAN OWN AND CROSS-PRICE ELASTICITIES[a]

| # | Brand | Corn Flakes | Frosted Flakes | Rice Krispies | Froot Loops | Cheerios | Total | Lucky Charms | P Raisin Bran | CapN Crunch | Shredded Wheat |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | K Corn Flakes | | | | | | | | | | |
| 2 | K Raisin Bran | | | | | | | | | | |
| 3 | K Frosted Flakes | | | | | | | | | | |
| 4 | K Rice Krispies | | | | | | | | | | |
| 5 | K Frosted Mini Wheats | | | | | | | | | | |
| 6 | K Froot Loops | | | | | | | | | | |
| 7 | K Special K | | | | | | | | | | |
| 8 | K Crispix | | | | | | | | | | |
| 9 | K Corn Pops | | | | | | | | | | |
| 10 | GM Cheerios | | | | | | | | | | |
| 11 | GM Honey Nut Cheerios | | | | | | | | | | |
| 12 | GM Wheaties | | | | | | | | | | |
| 13 | GM Total | | | | | | | | | | |
| 14 | GM Lucky Charms | | | | | | | | | | |
| 15 | GM Trix | | | | | | | | | | |
| 16 | GM Raisin Nut | | | | | | | | | | |
| 17 | GM Cinnamon Toast Crunch | | | | | | | | | | |
| 18 | GM Kix | | | | | | | | | | |
| 19 | P Raisin Bran | | | | | | | | | | |
| 20 | P Grape Nuts | | | | | | | | | | |
| 21 | P Honey Bunches of Oats | | | | | | | | | | |
| 22 | Q 100% Natural | | | | | | | | | | |
| 23 | Q Life | | | | | | | | | | |
| 24 | Q CapN Crunch | | | | | | | | | | |
| 25 | N Shredded Wheat | | | | | | | | | | |
| 26 | Outside good | | | | | | | | | | |

TABLE VII

MEDIAN OWN AND CROSS-PRICE ELASTICITIES[a]

| # | Brand | Corn Flakes | Frosted Flakes | Rice Krispies | Froot Loops | Cheerios | Total | Lucky Charms | P Raisin Bran | CapN Crunch | Shredded Wheat |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | K Corn Flakes | −3.379 | 0.212 | 0.197 | 0.014 | 0.202 | 0.097 | 0.012 | 0.013 | 0.038 | 0.028 |
| 2 | K Raisin Bran | 0.036 | 0.046 | 0.079 | 0.043 | 0.145 | 0.043 | 0.037 | 0.057 | 0.050 | 0.040 |
| 3 | K Frosted Flakes | 0.151 | −3.137 | 0.105 | 0.069 | 0.129 | 0.079 | 0.061 | 0.013 | 0.138 | 0.023 |
| 4 | K Rice Krispies | 0.195 | 0.144 | −3.231 | 0.031 | 0.241 | 0.087 | 0.026 | 0.031 | 0.055 | 0.046 |
| 5 | K Frosted Mini Wheats | 0.014 | 0.024 | 0.052 | 0.043 | 0.105 | 0.028 | 0.038 | 0.054 | 0.045 | 0.033 |
| 6 | K Froot Loops | 0.019 | 0.131 | 0.042 | −2.340 | 0.072 | 0.025 | 0.107 | 0.027 | 0.149 | 0.020 |
| 7 | K Special K | 0.114 | 0.124 | 0.105 | 0.021 | 0.153 | 0.151 | 0.019 | 0.021 | 0.035 | 0.035 |
| 8 | K Crispix | 0.077 | 0.086 | 0.114 | 0.034 | 0.181 | 0.085 | 0.030 | 0.037 | 0.048 | 0.043 |
| 9 | K Corn Pops | 0.013 | 0.109 | 0.034 | 0.113 | 0.058 | 0.025 | 0.098 | 0.024 | 0.127 | 0.016 |
| 10 | GM Cheerios | 0.127 | 0.111 | 0.152 | 0.034 | −3.663 | 0.085 | 0.030 | 0.037 | 0.056 | 0.050 |
| 11 | GM Honey Nut Cheerios | 0.033 | 0.192 | 0.058 | 0.123 | 0.094 | 0.034 | 0.107 | 0.026 | 0.162 | 0.024 |
| 12 | GM Wheaties | 0.242 | 0.169 | 0.175 | 0.025 | 0.240 | 0.113 | 0.021 | 0.026 | 0.050 | 0.043 |
| 13 | GM Total | 0.096 | 0.108 | 0.087 | 0.018 | 0.131 | −2.889 | 0.017 | 0.017 | 0.029 | 0.029 |
| 14 | GM Lucky Charms | 0.019 | 0.131 | 0.041 | 0.124 | 0.073 | 0.026 | −2.536 | 0.027 | 0.147 | 0.020 |
| 15 | GM Trix | 0.012 | 0.103 | 0.031 | 0.109 | 0.056 | 0.026 | 0.096 | 0.024 | 0.123 | 0.016 |
| 16 | GM Raisin Nut | 0.013 | 0.025 | 0.042 | 0.035 | 0.089 | 0.040 | 0.031 | 0.046 | 0.036 | 0.027 |
| 17 | GM Cinnamon Toast Crunch | 0.026 | 0.164 | 0.049 | 0.119 | 0.089 | 0.035 | 0.102 | 0.026 | 0.151 | 0.022 |
| 18 | GM Kix | 0.050 | 0.279 | 0.070 | 0.101 | 0.106 | 0.056 | 0.088 | 0.030 | 0.149 | 0.025 |
| 19 | P Raisin Bran | 0.027 | 0.037 | 0.068 | 0.044 | 0.127 | 0.035 | 0.038 | −2.496 | 0.049 | 0.036 |
| 20 | P Grape Nuts | 0.037 | 0.049 | 0.088 | 0.042 | 0.165 | 0.050 | 0.037 | 0.051 | 0.052 | 0.047 |
| 21 | P Honey Bunches of Oats | 0.100 | 0.098 | 0.104 | 0.022 | 0.172 | 0.109 | 0.020 | 0.024 | 0.038 | 0.033 |
| 22 | Q 100% Natural | 0.013 | 0.021 | 0.046 | 0.042 | 0.103 | 0.029 | 0.036 | 0.052 | 0.046 | 0.029 |
| 23 | Q Life | 0.077 | 0.328 | 0.091 | 0.114 | 0.137 | 0.046 | 0.096 | 0.023 | 0.182 | 0.029 |
| 24 | Q CapN Crunch | 0.043 | 0.218 | 0.064 | 0.124 | 0.101 | 0.034 | 0.106 | 0.026 | −2.277 | 0.024 |
| 25 | N Shredded Wheat | 0.076 | 0.082 | 0.124 | 0.037 | 0.210 | 0.076 | 0.034 | 0.044 | 0.054 | −4.252 |
| 26 | Outside good | 0.141 | 0.078 | 0.084 | 0.022 | 0.104 | 0.041 | 0.018 | 0.021 | 0.033 | 0.021 |

TABLE VIII

MEDIAN MARGINS[a]

| | Logit (Table V column ix) | Full Model (Table VI) |
|---|---|---|
| Single Product Firms | 33.6% (31.8%–35.6%) | 35.8% (24.4%–46.4%) |
| Current Ownership of 25 Brands | 35.8% (33.9%–38.0%) | 42.2% (29.1%–55.8%) |
| Joint Ownership of 25 Brands | 41.9% (39.7%–44.4%) | 72.6% (62.2%–97.2%) |
| Current Ownership of All Brands | 37.2% (35.2%–39.4%) | — |
| Monopoly/Perfect Price Collusion | 54.0% (51.1%–57.3%) | — |

- **Question**: What accounts for the differences in estimated margins between the Logit and Full Model?

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- **Question**: Why can Nevo compute margins for two additional models for logit but not for full model?